

Perform Network Enumeration on “scanme.nmap.org”. Find server IP then subnets os and devices info scan for the cve as well.

To perform the network enumeration on the target, we will use the tool “nmap”. First we will type the following command in the terminal of Kali Linux.

`nmap -A scanme.nmap.org`

```
(talha@kali)-[~]
$ nmap -A scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2023-11-08 07:55 EST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
|_ http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 30.97 seconds
```

Through this we found the server IP of our target. IP address is **45.33.32.156**

Finding subnets and OS detection:

To find the subnets and OS within the IP address range, we will use tool like “nmap”. It will be used with the “-v -A” flag. It will scans all the reserved TCP ports on the machine. The -v option enables verbose mode. The -A option will detect the OS. Use the following command:

`sudo nmap -v -A scanme.nmap.org`

```

(talha@kali)-[~]
$ sudo nmap -v -A scanme.nmap.org
[sudo] password for talha:
Starting Nmap 7.94 ( https://nmap.org ) at 2023-11-09 10:06 EST
NSE: Loaded NSE scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 10:06
Completed NSE at 10:06, 0.00s elapsed
Initiating NSE at 10:06
Completed NSE at 10:06, 0.00s elapsed
Initiating NSE at 10:06
Completed NSE at 10:06, 0.00s elapsed
Initiating Ping Scan at 10:06
Scanning scanme.nmap.org (45.33.32.156) [4 ports]
Completed Ping Scan at 10:06, 0.30s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 10:06
Completed Parallel DNS resolution of 1 host. at 10:06, 0.02s elapsed
Initiating SYN Stealth Scan at 10:06
Scanning scanme.nmap.org (45.33.32.156) [1000 ports]
Discovered open port 22/tcp on 45.33.32.156
Discovered open port 80/tcp on 45.33.32.156
Discovered open port 31337/tcp on 45.33.32.156
Discovered open port 9929/tcp on 45.33.32.156
Completed SYN Stealth Scan at 10:06, 10.12s elapsed (1000 total ports)
Initiating Service scan at 10:06
Scanning 4 services on scanme.nmap.org (45.33.32.156)
Completed Service scan at 10:08, 87.50s elapsed (4 services on 1 host)
Initiating OS detection (try #1) against scanme.nmap.org (45.33.32.156)
Retrying OS detection (try #2) against scanme.nmap.org (45.33.32.156)
Initiating Traceroute at 10:08
Completed Traceroute at 10:08, 3.37s elapsed
Initiating Parallel DNS resolution of 14 hosts. at 10:08
Completed Parallel DNS resolution of 14 hosts. at 10:08, 0.09s elapsed
NSE: Script scanning 45.33.32.156.
Initiating NSE at 10:08
Completed NSE at 10:08, 20.96s elapsed
Initiating NSE at 10:08
Completed NSE at 10:08, 2.02s elapsed
Initiating NSE at 10:08

```

Here it starts scanning the open ports and the OS on the target.

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Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 984 closed tcp ports (reset)

```

PORT	STATE	SERVICE	VERSION
9/tcp	filtered	discard	
13/tcp	filtered	daytime	
22/tcp	open	ssh	OpenSSH 6.6.1p1 Ubuntu 2ubuntu.12.1 (Ubuntu Linux; protocol 2.0)
ssh-hostkey:			
	1024	ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75	(DSA)
	2048	20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2	(RSA)
	256	96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57	(ECDSA)
	256	33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56	(ED25519)
25/tcp	filtered	smtp	
79/tcp	filtered	finger	
80/tcp	open	http?	
135/tcp	filtered	msrpc	
139/tcp	filtered	netbios-ssn	
445/tcp	filtered	microsoft-ds	
465/tcp	filtered	smtps	
587/tcp	filtered	submission	
593/tcp	filtered	http-rpc-epmap	
2525/tcp	filtered	ms-v-worlids	
4444/tcp	filtered	krb524	
9929/tcp	open	nping-echo	Nping echo
31337/tcp	open	tcpwrapped	

```

Aggressive OS guesses: Linux 2.6.32 (89%), Linux 3.4 (89%), Linux 3.5 (89%), Linux 4.2 (89%), Linux 4.4 (89%), Synology DiskStation Manager 5.1 (89%), WatchGuard Firewall 11.8 (89%), Linux 2.6.35 (88%), Linux 3.10 (88%), Linux 2.6.32 or 3.10 (88%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 35.845 days (since Wed Oct 4 14:51:25 2023)
Network Distance: 22 hops
TCP Sequence Prediction: Difficulty=263 (Good Luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

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TRACEROUTE (using port 53/tcp)
HOP RTT ADDRESS
1 3.20 ms 192.168.18.1
2 4.58 ms 202.163.72.253
3 5.25 ms 10.57.25.53
4 15.78 ms 182.184.3.149
5 ...
6 52.57 ms 10.253.4.22
7 ...
8 135.30 ms me1-b5-link.ip.twelve99.net (62.115.11.140)
9 ... 10
11 211.41 ms ash-b2-link.ip.twelve99.net (62.115.123.123)
12 220.96 ms akamai-ic-352275.ip.twelve99-cust.net (62.115.184.199)
13 211.46 ms ae2-r02-1ad01.icn.netarch.akamai.com (23.203.152.40)
14 ...
15 269.14 ms ae8-r01-sjc01.icn.netarch.akamai.com (23.32.63.47)
16 277.53 ms ae0-r02-sjc01.icn.netarch.akamai.com (23.32.63.75)
17 280.08 ms ae2-r12-sjc01.icn.netarch.akamai.com (23.207.232.41)
18 269.16 ms a23-203-158-53.deploy.static.akamaitechnologies.com (23.203.158.53)
19 ...
20 280.37 ms 10.203.35.71
21 ...
22 285.92 ms scanme.nmap.org (45.33.32.156)

```

```

NSE: Script Post-scanning.
Initiating NSE at 10:08
Completed NSE at 10:08, 0.00s elapsed
Initiating NSE at 10:08
Completed NSE at 10:08, 0.00s elapsed
Initiating NSE at 10:08
Completed NSE at 10:08, 0.00s elapsed
Read data files from: /usr/bin/./share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 131.22 seconds
Raw packets sent: 1144 (52.220KB) | Rcvd: 1048 (43.800KB)

```

After sometime, it scans all the active ports and OS on scanme.nmap.org. In the red border section, it scans the active TCP ports and their OS.

IP Address: 45.33.32.156

Open Ports: 9/TCP, 13/TCP, 22/TCP

Version: OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)

SSH-host key:

	1024	ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75	(DSA)
	2048	20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2	(RSA)
	256	96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57	(ECDSA)
	256	33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56	(ED25519)

All the open **PORTS**, **STATE**, **SERVICE** and **VERSION** are highlighted in the red box of screen shots.

Compile the brief document of exploitation use CVE-2016-0012

Microsoft Office ASLR Bypass - CVE-2016-0012

An attacker who successfully exploited it could bypass the Address Space Layout Randomization (ASLR) security feature, which helps protect users from a broad class of vulnerabilities. However, an attacker could use this ASLR bypass in conjunction with another vulnerability, such as a remote code execution vulnerability, to more reliably run arbitrary code on a target system.

In a web-browsing scenario, successful exploitation of the ASLR bypass requires a user to be logged on and running an affected version of Microsoft Office. The user would then need to browse to a malicious site. Therefore, any systems where a web browser is used frequently, such as workstations or terminal servers, are at the most risk from this ASLR bypass. Servers could be at more risk if administrators allow users to browse and read email on servers. However, best practices strongly discourage allowing this. The update addresses the ASLR bypass by helping to ensure that affected versions of Microsoft Office properly implement the ASLR security feature.

CVE-2016-0012 is a Common Vulnerabilities and Exposures (CVE) identifier, which is a unique identifier assigned to a security vulnerability.

Summary of CVE-2016-0012

Microsoft Office 2007 SP3, Excel 2007 SP3, PowerPoint 2007 SP3, Visio 2007 SP3, Word 2007 SP3, Office 2010 SP2, Excel 2010 SP2, PowerPoint 2010 SP2, Visio 2010 SP2, Word 2010 SP2, Office 2013 SP1, Excel 2013 SP1, PowerPoint 2013 SP1, Visio 2013 SP1, Word 2013 SP1, Excel 2013 RT SP1, PowerPoint 2013 RT SP1, Word 2013 RT SP1, Office 2016, Excel 2016, PowerPoint 2016, Visio 2016, Word 2016 and Visual Basic 6.0 Runtime allow remote attackers to bypass the ASLR protection mechanism via unspecified vectors, aka "Microsoft Office ASLR Bypass."

The above mentioned products were affected by the CVE 2016-0012.